StoryScope: what the paper actually found

A reading of Russell, Rajendhran, Pham, Iyyer & Wieting, arXiv:2604.03136 (v6, 10 Aug 2026)

Everything below is taken from the paper itself (abstract page + full HTML). No numbers are invented. Where the paper reports a number, it is quoted; where it reports a mechanism, it is paraphrased.

1 The question

Most AI-text detectors look at surface style: word choice, sentence rhythm, the em-dash problem, words like "delve" and "tapestry". Those cues work, but they are fleeting. The authors point out that GPT-5.4 already cut em-dash usage sharply, and that fine-tuning an LLM to mimic human style drops detection on creative writing from 97% to 3% (Chakrabarty et al. 2026). If the surface is this easy to change, detection built on the surface will not last.

So the paper asks a different question: can a story be identified as human or AI from *discourse-level narrative choices* alone—plot structure, character agency, temporal ordering, information revelation—with style stripped out entirely? Changing narrative structure is not a post-hoc edit; it is a rewrite. A detector built on narrative features should survive the edits that kill style-based detectors.

2 The method

Corpus. 10,272 human-written short stories from Books3. Each was reverse-engineered into a writing prompt (Gemini 2.5 Flash inferred the premise), and five LLMs—Gemini 3 Flash, Kimi K2.5, DeepSeek V3.2, Claude Sonnet 4.6, GPT-5.4—each wrote a story from the same prompt. Six sources total, 61,608 stories, mean 4,753 words. Source identities were anonymized in all LLM-facing prompts.

Pipeline. Three stages:

1. *Structured templates.* Each story is converted by GPT-5.1 into a template organized along ten narrative dimensions from the NarraBench taxonomy: Agent, Social Network, Event, Plot, Structure, Setting, Time, Revelation, Perspective, Style. Prose becomes fields: character names and motivations, causal chains, key events, temporal order.

2. *Cross-source comparison.* A held-out pool of 600 stories (100 prompts, all six versions) is presented in template form to GPT-5.1 with high reasoning effort, producing structured comparative analyses: per-source notes, cross-source divergences, recurring patterns. This compresses ∼2.7M tokens of raw text into ∼686K tokens.

3. *Feature discovery.* The analyses yield 408 candidate features, each a question with typed answers (categorical, ordinal, scale, binary, multi-select). Deduplication by embedding clustering (F2LLM-4B, single linkage at cosine 0.85) gives **304 features**. A style-dependence audit excludes 47 style-linked features, leaving **257 strict narrative features** for the main experiments.

Classifier. XGBoost (beat linear models and random forests). Split: 7,383 train / 1,405 val / 1,384 test prompts; tuned on validation, retrained on train+val, reported on test (8,301 stories).

Reliability. Production extractor (Gemini 3 Flash): Krippendorff's $\alpha = 0.90$ over five runs, mean pairwise Cohen's $\kappa = 0.89$. Human validation over 12 stories (240 items per annotator): human-vs-model $\kappa = 0.84$, versus human-vs-human 0.74. The extractor agrees with humans at least as well as humans agree with each other.

3 The numbers

Binary human-vs-AI detection (macro-F1):

Method	Features	F1	AUPRC
Narrative (strict, no style)	257	**93.2**	.959
Core only	30	84.8	.828
Core + fingerprint	101	91.1	.934
Narrative + Style	304	96.0	.982
Style only	39	85.8	.867
ModernBERT (raw text)	—	99.9	1.00
Stylometric + XGB	144	99.8	.999
TF-IDF + XGB	5,000	99.7	.999
Binoculars (zero-shot)	—	55.9	.404

Narrative features alone get 93.2%—2.8 points below narrative+style, retaining over 97% of the full model's performance. A compact 30-feature core keeps 84.8%.

Six-way authorship attribution (macro-F1):

Method	F1
Narrative	**68.4**
Core only	46.5
Core + fingerprint	63.4
Narrative + Style	77.3
Style only	60.4

Narrative beats style-only by 8.0 points (CI 6.7–9.2). Attribution is much harder than detection; the AI models overlap heavily with each other.

Robustness to editing. 278 Gemini stories were run through LAMP (a span-level rewriter removing seven categories of AI artifacts, using Gemini as the rewriter). The narrative classifier still detects edited stories at **93.9%** macro-F1 (AUPRC .988) versus 95.5% on originals: a 1.6-point drop.

4 What separates human from AI, in narrative terms

- **AI over-explains its themes.** ~20% more explicit and moralizing on a 1–5 scale. Narrators state the theme 77% of the time vs. 52% for humans. AI dialogue serves philosophical debate 59% vs. 34%. Allusions are vague (72% vs. 50%). The meaning is said, not inferred.

- **Human authors subvert linearity.** AI has tighter causal chains, more protagonist-driven resolutions (69% vs. 46%), fewer subplots (79% "no subplots" vs. 57%). AI resolutions favor internal acceptance (47% vs. 27%); humans tolerate ambiguity and use time jumps, flashbacks, nonlinear structure. A human mystery opens at the funeral and spirals backward; AI tells it from first clue to grand reveal.
- **AI over-writes the body and senses.** Emotion through physical sensation 81% vs. 38%, smell imagery 82% vs. 57%. Humans name feelings: explicit emotion labels 29% vs. 8%. Where a human writes "she felt afraid", AI writes a tightening chest, cold sweat, dimming lamplight.
- **Human authors engage the outside world.** Specific text/author references at nearly double the AI rate (47% vs. 24%), balanced explicit/implicit references (37% vs. 16%), fourth-wall breaks 67% vs. 39%, direct reader address 28% vs. 7%. AI writes as though no one is watching.
- **AI has less diverse narrative features overall.** Humans span more locations, carry more dialogue, integrate more subplots (42% vs. 21%), give protagonists moral ambivalence more often (59% vs. 38%).

5 The per-model fingerprints

- **Claude keeps it cool.** Most distinctive profile. Restraint: flat event escalation, most uniform voice. Reverent toward literary tradition (62% vs. 39–56%). Favors epilogues, avoids dream sequences, prefers quiet endings.
- **GPT likes to gossip.** Gossip/rumor as plot mechanism (64% vs. 44–55%), stories as reflection on events years ago, ensemble social networks. Subverts expectations more (41% vs. 27–36%), leaves reconciliations ambiguous.
- **DeepSeek front-loads.** Puts crucial context early that other sources save.
- **Gemini, tidiest and bleakest.** Tidiest endings, extended denouements, bleakest settings (88% bleak and oppressive).
- **Kimi is the generic center.** Fewest fingerprints, lowest attribution F1.

AI convergence. The five AI sources cluster in a shared region of narrative space, well separated from humans. Human-AI centroid distance is 1.6× the AI-AI distance; even the closest human-AI pair is farther than the most distant AI-AI pair. The six most-confused attribution pairs are all AI-vs-AI; the largest (Gemini $\leftrightarrow$ DeepSeek, 222 and 207 stories) dwarfs the most common human misclassification (Human $\rightarrow$ Kimi, 46).

Human narratives are rarer. Rarity = mean Euclidean distance to 25 nearest neighbors. Humans: mean rarity percentile 0.71 vs. 0.49 (Cohen's $d = 0.83$). 24.7% of human stories in the top-10% rarest versus 7.1% of AI (top 1%: 3.0% vs. 0.6%). At the prompt level, the human version is the rarest of the six 57.8% of the time (chance: 16.7%). Human stories are also more dispersed (22% larger centroid radius, 1.13× larger median 10-NN radius).

6 What it means

The durable claim is not "we can detect AI with 93% F1"—raw-text detectors do 99.9%. It is that *narrative structure itself carries the signal, independent of style*, and that this signal survives the edits that defeat style detectors. Changing the narrative choices means rewriting the story. So narrative features are a more durable basis for authorship analysis, and the rarity measure gives a concrete, measurable proxy for originality.

There is a flip side the paper mostly leaves implicit: the same feature space is a recipe list for making AI fiction *less* detectable. Over-explain less, break the causal chain, add subplots, name real books, address the reader, let endings dangle, give the protagonist a moral mess. And since humans are rarer and more dispersed, the gap will shrink as models are pushed toward the human region of narrative space.

7 Method caveats

- Books3 short stories, mean 4,753 words. The pipeline needs long texts; transfer to flash fiction, poetry, or non-fiction is untested.
- Human stories are one per prompt and *generated* the prompt; AI stories are mirrors. The comparison is human-original vs. AI-mirror, not equal footing.
- All feature extraction uses LLMs (GPT-5.1 templates, GPT-5.4 style audit, Gemini 3 Flash production). Extractor-vs-human $\kappa = 0.84$ is good but not perfect; the feature space is partly an LLM's view of narrative.
- The paper uses Books3 (copyright-problematic dataset; academic use stated). The AI disclosure notes Claude Code and Codex were used to write and polish the paper itself.